

ABSOLUTE AMPLITUDES AND SPECTRA OF CERTAIN MUSICAL INSTRUMENTS AND ORCHESTRAS¹

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In a paper on "Speech Power and its Measurement,"² one of the authors has given some measurements of average and peak amplitude in speech, using apparatus in which the speech spectrum was divided into thirteen bands of frequencies. The same apparatus has been used in a series of measurements on musical instruments, which are reported in this paper.

As with the speech measurements, the data are statistical in nature, and are taken with a view to their engineering applications. These applications are concerned, chiefly, with the transmission and reproduction of music, and the data should show the power and frequency requirements for systems which are called upon to perform these functions without distortion. In carrying out this purpose it has been thought well to measure both individual instruments, and instruments playing together in orchestras; to make measurements on actual musical selections, rather than on single notes; and to take the measurements in such a way as to obtain an average or integrated picture of the selection, as well as the distribution of amplitudes in magnitude and frequency, the extreme values being particularly important.

APPARATUS

The apparatus was developed for the purpose of making just this sort of statistical measurement on sounds. Since it was not described in the paper referred to above, a brief description will be given here.

A block diagram is shown in Fig. 1. The fourteen band-pass filters were designed so that when connected in parallel on the input side, and with the output of each filter terminated in a 600-ohm resistance, they present a uniform impedance of 600 ohms at all frequencies. Only one filter at a time is used, this one being connected to the 600-ohm measuring circuit, while all the others are connected to 600-ohm resistances. Fig. 2 shows a typical filter attenuation characteristic. The ranges of frequencies covered by the filters may be obtained from the curves showing the data obtained.

¹ Presented before Acoustical Society of America, December 14, 1929

² Bell System Technical Journal, October, 1929.

Two distinct types of measurement are made. The "average amplitude meter" consists of an amplifier, a rectifier, and a shunted flux-

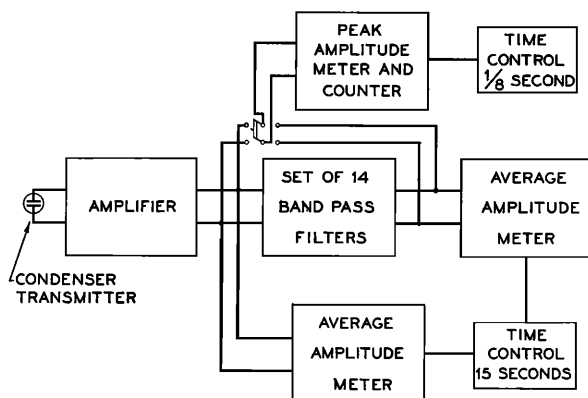


FIG. 1. Diagram of circuit for measuring peak and average amplitudes in music.

meter. The rectifier is a two-element vacuum tube with a resistance in series, of such a value that within the range of amplitude expected it is

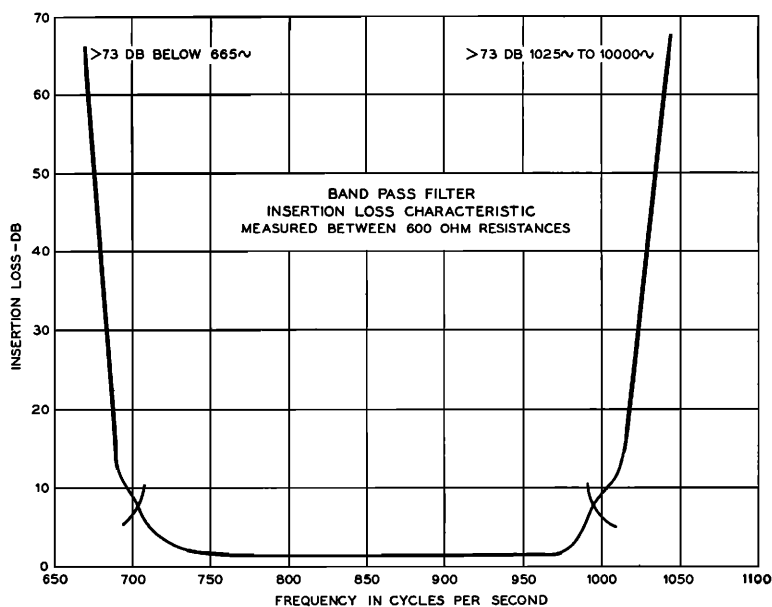


FIG. 2. Attenuation curve of 700-1000 c. p. s. band-pass filter. Lines at right and left show where curves for adjacent filters cross. Attenuations shown are typical of all filters used.

very nearly linear in response. Fig. 3 shows a curve of the output against a sine wave input, the limits of actual use being marked. The

dotted line shows the true linearity which the curve approaches. The possible error at the lowest output is 1.5 decibels, but this rapidly diminishes as the output increases.

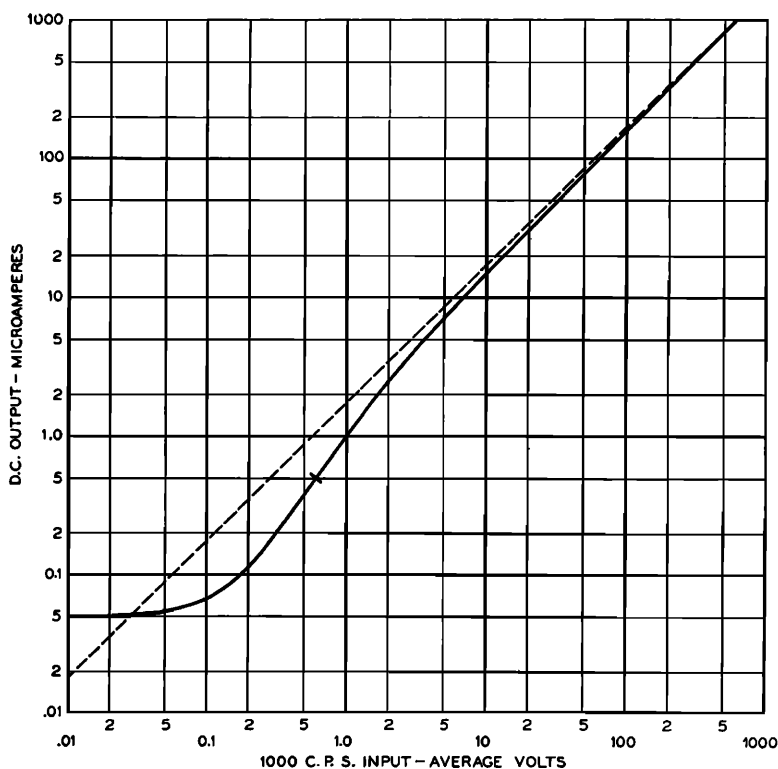


FIG. 3. *Response of two-electrode, vacuum-tube rectifier, with 300,000-ohm resistance in series. Cross mark shows lowest average output used in measurements, while dotted curve shows the characteristics of a perfect rectilinear rectifier.*

The indicating instrument in the average amplitude meter is a Gras-sot fluxmeter. This is a very convenient instrument for measuring quantity of electricity over periods much greater than those in which a ballistic galvanometer may be used. It is essentially a galvanometer in which the restoring torque is made very small, yet which has a high electromagnetic damping when placed in a low resistance circuit. Under these conditions the needle moves at a rate proportional to the current and remains almost stationary when the current stops. In a vacuum tube circuit the low resistance required for damping must be provided by means of a shunt. The sensitivity is proportional to the resistance of

the shunt, but a limit is set by the fact that the torque cannot be reduced entirely to zero, and results in a drift of the needle with a rate proportional to the sum of the resistances of shunt and meter. The limiting effect of this drift is not directly upon the quantity sensitivity, but rather upon the relation of current and time for a given sensitivity, the maximum allowable error having been established. Thus, with the meters used, the resistance being 23.5 ohms, it was found that with a shunt of 24.5 ohms the drift in 15 seconds would not be greater than 5 per cent of the deflection from the natural point of rest. If 5 per cent is set as the limit of error, greater intervals than 15 seconds could not be used without decreasing the sensitivity by lowering the shunt; and greater sensitivity could be used only by shortening the interval. A final limit to the length of interval that may be used, for a given percentage error, is set by the resistance of the meter itself. With the meters used this limit is about half a minute for a 5 per cent error. In all of our work the 15-second interval has been used. In order to make use of the entire scale with equal percentage error for all deflections, the natural point of rest is adjusted to one end of the scale. Provision is made for returning the needle to this point electrically after reading a deflection. Calibration of the meters is made with constant currents over 15-second intervals, so that errors in reading due to drift arise only when the distribution of current over the interval is very irregular.

It is seen that the fluxmeter, used with the linear rectifier over a definite time interval, shows the average amplitude during the interval. Two of the average amplitude meters are used, the "total" responding to all frequencies, the "band" to those passing the filter in use. Measurements with both meters are made regularly in alternate 15 second intervals, the intervening intervals being used for reading and resetting the fluxmeters. The range covered in the band average amplitude meter is 35 db., and in the total 33 db. This is accomplished in the first case by using the mirror attached to the fluxmeter coil, for very small deflections; and in the other case by using two fluxmeters in series, one with a smaller shunt than the other, and read only when the other meter is off scale.

The other type of measurement is that of peak amplitudes. The most important part of the "peak amplitude meter" is a set of gas filled vacuum tubes.³ In a tube of this kind, with a given plate voltage and

³ This is a special three-electrode, hot-cathode tube filled with argon, constructed in the Bell Telephone Laboratories. A relay tube of this general type was first devised by A. W. Hull, —e.g. General Electric Review, April and July, 1929.

grid bias, a peak voltage of a certain definite magnitude applied to the grid causes an arc to strike between plate and cathode, and this current then continues to flow. It is large enough to operate a sensitive relay, being of the order of 10 or 20 milliamperes. The arrangement of these tubes is shown in Fig. 4. Ten tubes are used, divided into two groups of five each. These groups are fed by amplifying tubes, one of which is set

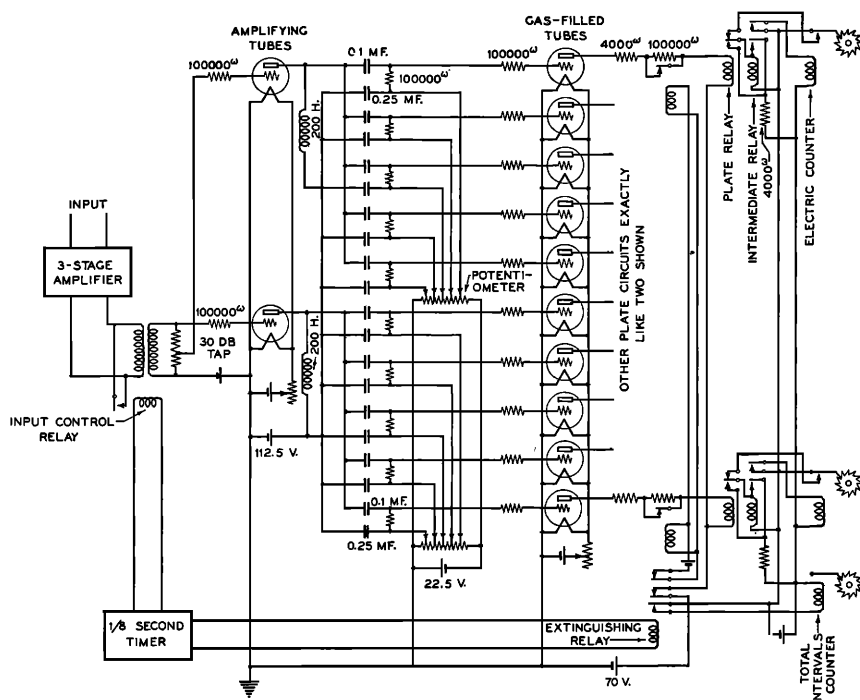


FIG. 4. Diagram of peak-amplitude meter. Peak amplitudes in alternate one-eighth second intervals are measured and counted over a period of any desired length.

at a level 30 decibels above the other. Within a group of five tubes, the grid biasing voltages are adjusted so that the peak voltages required to strike the different tubes are progressively higher. The difference between successive tubes is set at 6 decibels, so that the entire range covered between first and tenth tube is 54 decibels. A relay is shown in the plate circuit of each tube. These relays operate electrical counters, which count each flash of the corresponding tubes. In order to insure a count even on a very brief flash, intermediate relays are used which hold themselves closed until the counter armature has completed its stroke, at which point it makes a contact; also, to prevent repeated

counting, the intermediate relay does not release until the first relay has fallen back, the back contact of the first relay and the contact of the counter being connected in series to release the intermediate relay. The tubes are extinguished by means of a relay opening the common plate circuit. In reconnecting, however, in order not to flash all tubes at once, the precaution must be taken of introducing 100,000 ohms into each plate circuit, and shorting it out again when the connection has been made. Other means of extinguishing may be used, as for instance the

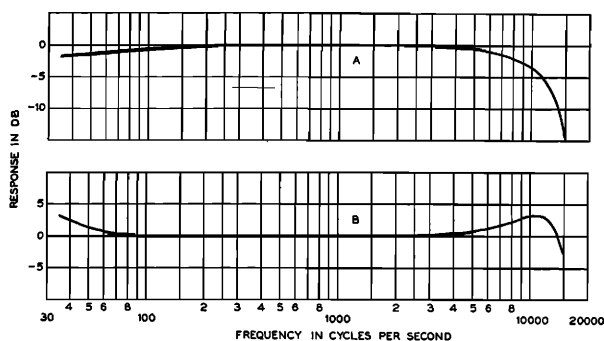


FIG. 5. *Frequency characteristics of electric circuit, giving relative outputs for a constant input* Curve A includes the "average amplitude meter," while curve B includes the "peak amplitude meter."

brief connection of a condenser around the plate circuit resistance of each tube; but the method used was found to give the most reliable results in a very short interval.

In taking measurements with the peak amplitude meter, the tubes are connected to the circuit for one-eighth second, and the highest tube striking indicates within 6 db. the highest peak during that interval. Each tube striking records the fact by operating its relay and counter. The tubes are then extinguished during the next one-eighth second, then reconnected to the circuit. Peak measurements are made in this way during alternate one-eighth second intervals, for as long a time as desired. The counter readings at the end show how many times each tube flashed, and by a subtraction of numbers for consecutive tubes, the number of peaks lying in each 6 db. zone is obtained. As is indicated in Fig. 1, the peak amplitude meter may be connected either to the input or to the output side of the filters.

Time controls for the 15-second and one-eighth-second intervals are provided by means of synchronous motors. These drive switches which operate relays to perform the desired functions. The various amplifiers

in the circuit are provided with variable gain, for adjustment to different sound levels. Frequency characteristics of the circuit are shown in Fig. 5. These include all electric parts of the circuit, but not the pressure-voltage characteristics of the condenser transmitters. The latter, obtained by means of a thermophone calibration,⁴ are shown for two transmitters in Fig. 6. The transmitter No. 2 was used only for the bands above 8000 c.p.s., where it has greater sensitivity than the first transmitter.

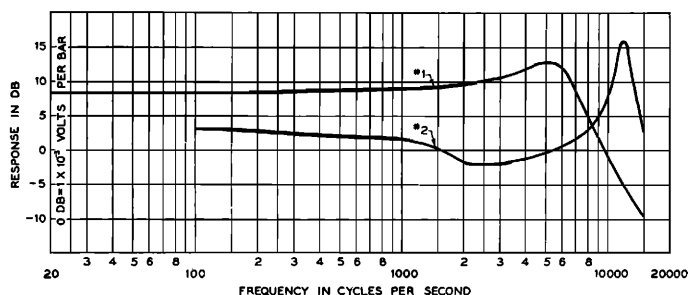


FIG. 6. Pressure-voltage calibrations of the two condenser transmitters.

METHODS OF OBTAINING DATA

The individual instruments chosen for measurements were chiefly those believed to be possible extreme cases for reproduction, either in frequency range or in absolute amplitude. The list is as follows: bass drums of three sizes (29"×12", 36"×15", 34"×19"), snare drum, cymbals, triangle, bass viol, bass saxophone, bass tuba, trombone, trumpet, French horn, clarinet, flute, piccolo, piano, and pipe organ. In addition measurements have been made upon orchestras of fifteen, eighteen, and seventy-five pieces.

With the exception of the pipe organ and seventy-five piece orchestra, the instruments were located in a large sound-proof laboratory room, 29'×29'×13' in size, and having a reverberation time of approximately one second for frequencies between 60 and 4000 c.p.s. The condenser microphone was placed at a distance of 10 ft. or more from the nearest wall and at a height from the floor about equal to that of the instrument under test, varying from about 2 ft. for the bass drum to about 5 ft. for the tuba. The instrument, in most cases, was placed 3 ft. in front of the microphone, though in some cases it was as much as 5 ft., and for the

⁴ See "The Absolute Calibration of Condenser Transmitters" by L. J. Sivian, Bell System Technical Journal, January, 1931.

piano 10 ft. With the orchestras of fifteen and eighteen pieces, the microphone was placed near the conductor, its distance from the different pieces varying from 6 to 20 ft., with the violins closest. The measuring apparatus, except for the microphone and one stage of amplification, was located in a separate room.

The organ and seventy-five piece orchestra were picked up in a large motion picture theater. For the orchestra the regular theater programs were used, but the organ was played particularly for the tests. The microphone was fastened to the conductor's stand, where it served to pick up both orchestra and organ, since the latter opened directly into the orchestra pit.

Only professional musicians were employed for the tests. In each case the musician was instructed to play a selection representative of his instrument, usually including extremes of range and volume, and different types of playing. Selections were made at least two minutes long, so that at least four average amplitude readings and at least 480 peak amplitude readings were obtained during the selection.

While it would be desirable to make measurements in all frequency bands during one playing of a selection, this would require a large multiplication of apparatus and observers, and was not found to be practicable. The selection was therefore repeated, once to obtain the distribution of peaks in the whole spectrum, and once for each of the frequency bands. In all cases readings of total average amplitude were taken simultaneously with the peak and average band amplitudes. Musicians were instructed to make repetitions as nearly alike as possible. In almost all cases they sounded closely alike to the ear, and in some cases the musician was able to make the fifteen repetitions with a total variation in the total average amplitude of only 2 db. In the case of the piano, a reproducing mechanism was used, and the selection could be repeated at will. The orchestras were not under our control, and the curves shown for them are not for the same selection in all bands. However, for the smaller orchestras all selections used were similar, in that they covered a large frequency range and used all instruments most of the time. For the theater orchestra, use was made of a number of repetitions of the same program during a week, and all bands presented together in our curves are at least for very similar types of music.

REDUCTION OF DATA

Calibration of the entire electrical circuit is accomplished by means of sine wave voltages, introduced into the circuit in series with the con-

denser transmitter. The r.m.s. voltage of the sine wave is measured by means of a vacuum thermocouple, followed by a known amount of attenuation. The r.m.s. value may be reduced to peak or average, as desired. The calibration is made in such a way that, for all frequency bands and all gain adjustments, the peak input required to strike any tube of the peak amplitude meter is known, and the average input corresponding to any 15 second fluxmeter deflection is also known. With these calibrations, it is believed that the error involved in measuring the average voltage generated in the circuit by sound falling upon the diaphragm of the transmitter is not greater than 2 decibels. While the voltage just required to strike a given tube of the peak amplitude meter is accurately known, a single peak measurement must be uncertain to the extent of the six decibel interval between tubes. When, however, a large number of measurements are made, and the number falling in one six decibel zone is compared with the numbers in the zones on either side, the actual distribution of peak amplitudes within the zone may be fairly accurately estimated. This will be more readily seen when the curves of data obtained are examined.

A greater uncertainty is involved in converting these voltages into sound pressures. In the first place, the frequency characteristic of the apparatus is much less regular when that of the transmitter is included. A rapidly changing characteristic within a frequency band makes it impossible to choose a calibrating frequency for that band which will give accurate results for all possible distributions of sound amplitudes. This is more particularly the case with the bands above 5000 c.p.s. However, the errors due to this cause are not likely to amount to more than a few decibels, and in most cases are much less than this.

In the second place, while the thermophone calibration of the transmitter, given in Fig. 6, shows accurately the sound pressure on the diaphragm for a given voltage measured in the circuit, this is not the pressure that would exist in the sound field at the same point were the transmitter absent. We have corrected our results for this distortion of the field, using measurements made at various frequencies and angles of incidence, by Messrs. R. K. Bonell and E. M. Little, of these laboratories. These measurements were made with a transmitter having the same geometry as our own. In these corrections normal incidence of the sound waves on the transmitter diaphragm was assumed. This is a correct assumption only when combined with further corrections explained in the following paragraph.

A third uncertainty in the conversion of measured voltages to sound

pressures arises in connection with reflections of sound waves within the room. While the walls and ceiling were covered with sound-absorbing material, still it was found that reflections began to have an appreciable effect when the source was two feet or more distant from the transmitter. As all our measurements were made at distances of three or more feet, the effect of reflections must be considered. These reflections distort the spectrum, both by selective absorption of the walls, and by the introduction of sound waves striking the diaphragm at other angles than normal. They also tend to raise the average total sound pressure. We have determined corrections for all of these effects at once by a series of auxiliary measurements, in which a complex sound source of constant composition was used. This was placed first at a distance of one foot, directly in front of the transmitter, and an analysis of the sound pressures made, using the total and band average amplitude meters. At this distance it is safe to assume that reflections had no appreciable effect. The source was then moved to other distances and the measurements repeated. The changes found in different parts of the spectrum were due to reflections, and the same changes could be assumed to take place with musical instruments at the same distances. To take care of instruments having different directional properties, the complex source, a loud speaking receiver, was equipped with horns of different sizes. A 1/2" tube was used as a point source, then horns with openings to 6-7/8" and of 22". The measurements with these horns were applied of the trombone and tuba, respectively. For the piano a special set of measurements was made, using the piano itself as the source. A 15-second section of a piano record was used, and measurements made at different distances, care being taken to include exactly the same section of the roll in each playing. For orchestras playing in the laboratory room, there is no definitely assignable distance. Measurements were made with the complex point source at a distance of 10 ft. in front of the transmitter, and also at 12 ft., 90° from the normal to the diaphragm. The two results are nearly alike, indicating that at these distances the sound from a non-directional source becomes almost completely diffused. An average of the two results was taken and used for correcting the orchestra data.

The net result of the two corrections explained in the preceding two paragraphs is to give a spectrum of the sound pressures that would exist in the field with no transmitter present and with no reflecting surfaces near. While these corrections have been found on the basis of average amplitude measurements only, the same corrections are assumed to hold for peak amplitudes.

For measurements in the theatre, the corrections for reflection of waves have not been directly determined. That there were appreciable

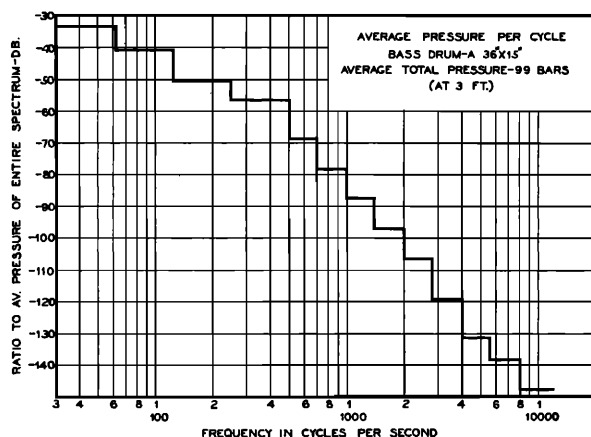


FIG. 7.

reflections seems certain. This is more particularly true for the organ, for which the orchestra platform was lowered to the bottom of the pit, and a semi-circular reflecting wall about ten feet high thereby intro-

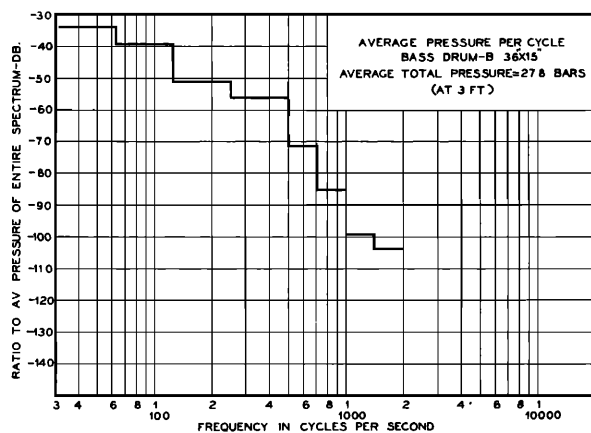


FIG. 8.

duced behind the transmitter. In addition, no audience was present during the organ tests. The best available data would seem to be those used for correcting the orchestra music in the laboratory room; hence these same corrections have been applied to the orchestra and organ in the theater.

Another possible source of error is that in some way an output may be obtained in certain bands where no frequency within these bands is present in the original sound. This might arise from modulation due to non-linearity of some part of the system, or it might arise from lack of sufficiently high attenuation in the filters, for frequencies outside their transmitted bands. A test has been made in which either of these conditions would have been discovered if it existed. A very pure wave was introduced into the system, and the outputs in all other bands compared with that in the band containing this frequency. The peak amplitude of the wave was made equal to the largest peak amplitudes found in the music measurements. The results showed definitely that no account need be taken of this effect in any of our music data.

METHODS OF PRESENTING DATA

The results of the measurements and corrections are shown for average amplitudes in Fig. 7 to 32, and for the peak amplitudes in Fig. 33 to 58.

The average amplitude curves are plotted in the following manner: The four or more average amplitude readings taken on a given band are averaged, as are also the readings of the total average amplitude taken simultaneously. The figure obtained for the band is divided by that for the total, and also by the width of the band in cycles per second. The result, expressed in decibels, is the ordinate for that band, and it is drawn on the curve as a straight line extending through the band. It represents the pressure per cycle in the band, referred to the total average pressure as zero level. The average of all values obtained for the total average pressure is given along with the curve, so that absolute values in each band may be calculated if desired. It should be pointed out that the lower limit of the first band is taken at 30 cycles, while the upper limit of the thirteenth band is taken at 12000 cycles. These points are chosen as practical limits set by the apparatus, in connection with the rate at which sound pressures are falling off. What we have called band 13-A, however, has a definite upper limit at 11300~, and in some of the curves this limit is shown. It should also be mentioned that, while the relative ordinates in the different bands have been corrected for reflections in the room, as explained above, the absolute value of the total average amplitude given with the curve is not the corrected value, but is the actual measured value obtained in the room at the distance given.

Peak amplitudes are presented in a different manner. The first curve

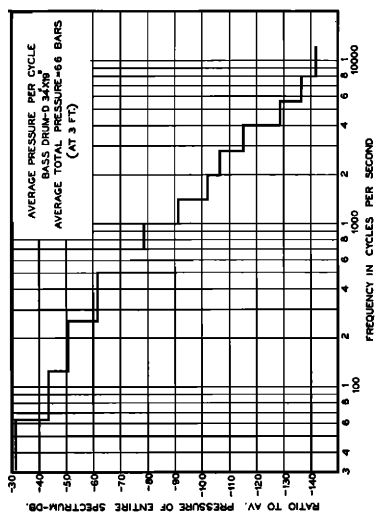


FIG. 10.

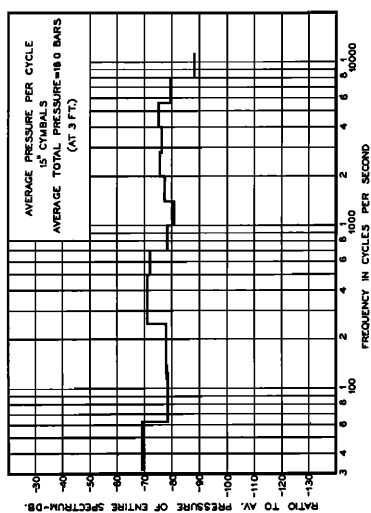


FIG. 12.

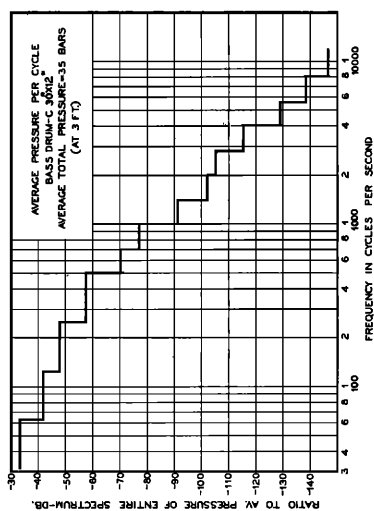


FIG. 9.

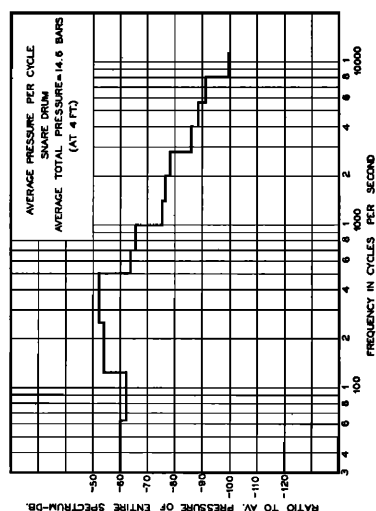


FIG. 11.

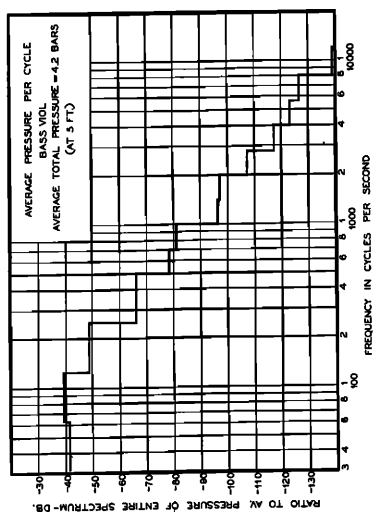


FIG. 14.

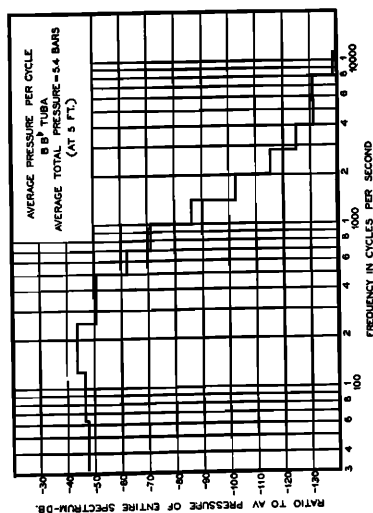


FIG. 16

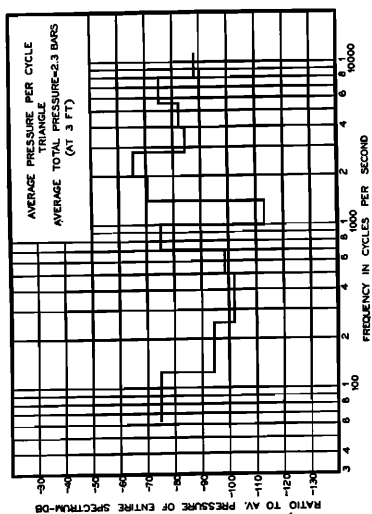


FIG. 13.

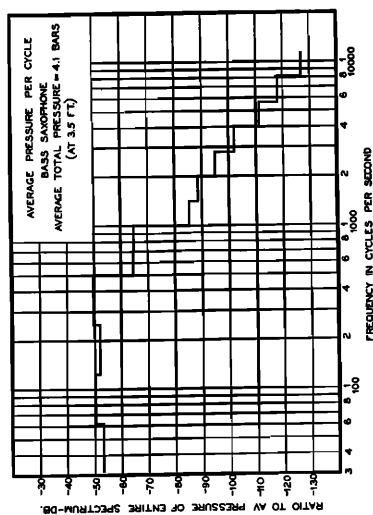


FIG. 15.

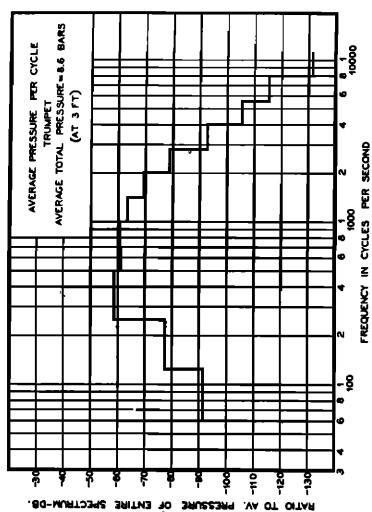


FIG. 17.

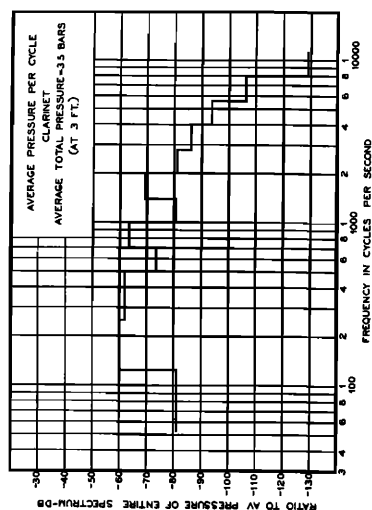


FIG. 18.

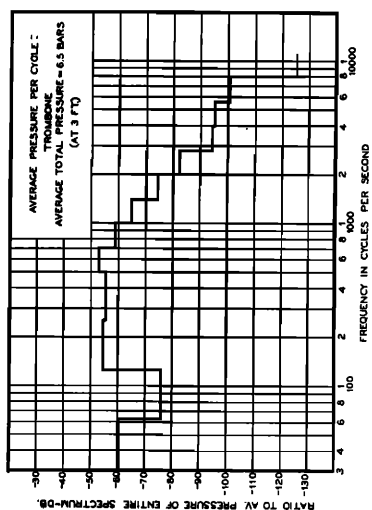


FIG. 19.

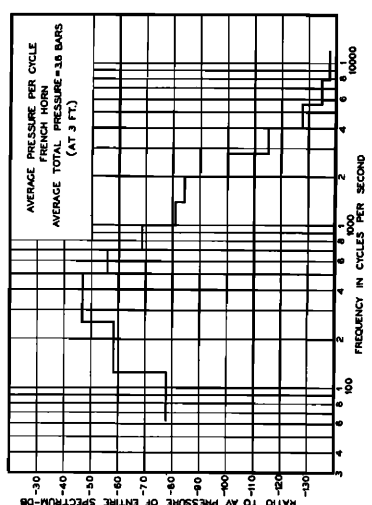


FIG. 20.

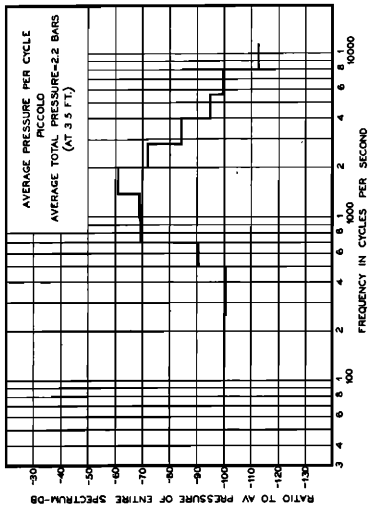


FIG. 21.

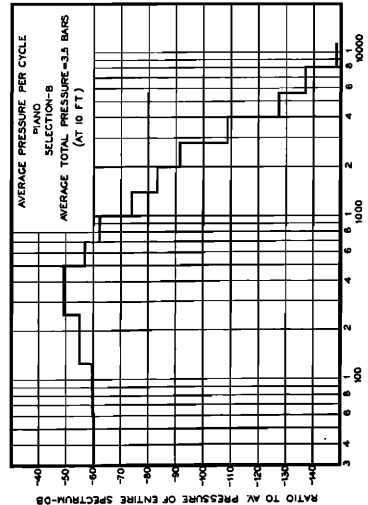


FIG. 22.

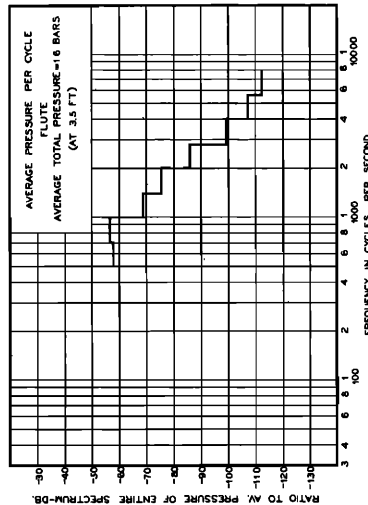


FIG. 23.

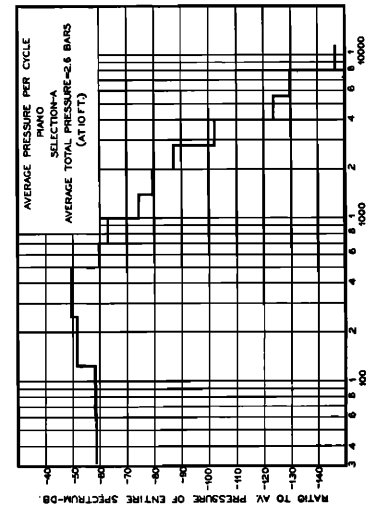


FIG. 24.

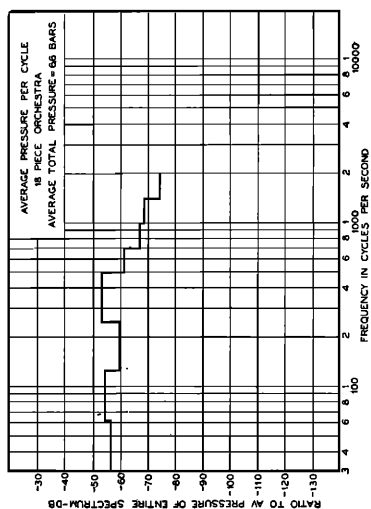


FIG. 26.

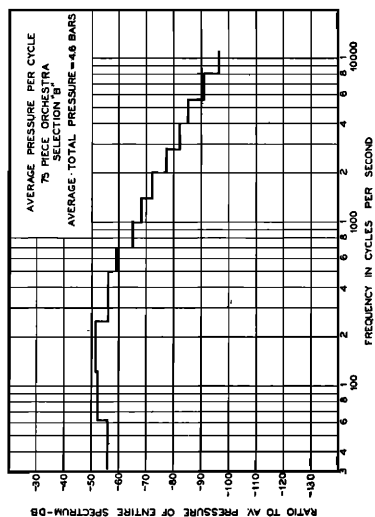


FIG. 28.

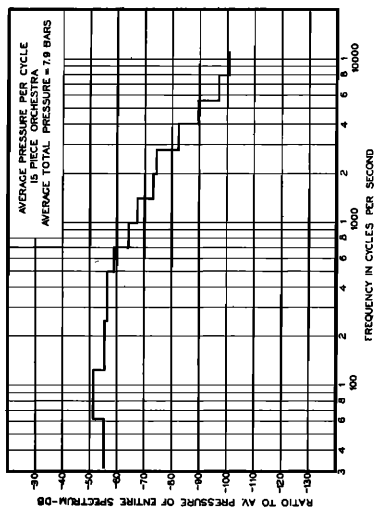


FIG. 25.

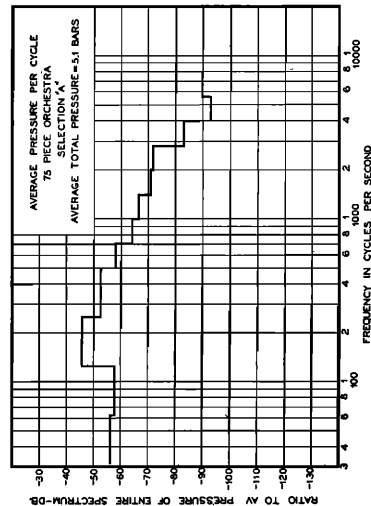


FIG. 27.

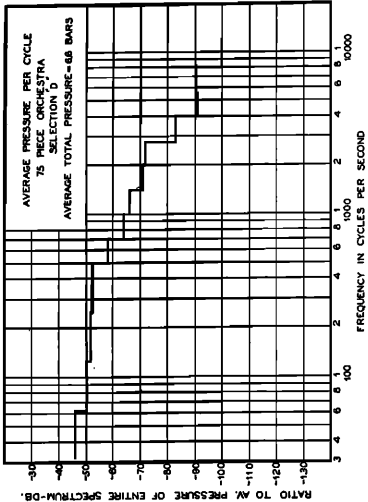


FIG. 29.

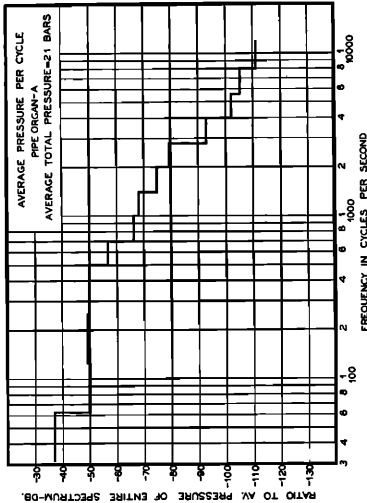


FIG. 31.

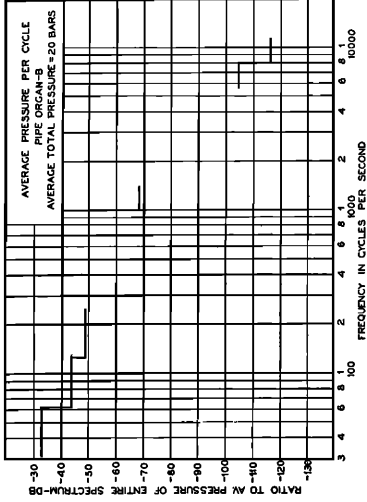


FIG. 30.

FIG. 32.

in each group shows the distribution of "total" peaks in the different amplitude zones, and the other curves give similar data for the different bands of frequencies. The peaks are measured in amplitude zones 6 db. wide. The number falling in a given zone is reduced to percent of the total number of measurements, and this divided by six, the width of the zone. This gives the abscissa of that section of the curve, which is drawn as a straight line extending through the range of amplitudes covered by the zone. The total average amplitude obtained at the same playing of the selection is used as zero level, and the ordinates of the curve are pressures above and below that value, expressed in db. Dividing the percentages by the zone widths gives a curve whose integral over all amplitudes is just 100 percent. A smooth curve might be drawn through the steps in such a way as not to change the integral under each section, and such a curve would approximate the true distribution of peaks over all amplitudes. As with the average curves, absolute values may be calculated by applying the total average pressure given with the curves. In some cases the curves are left open at the bottom, indicating that an appreciable percentage of the peaks lay below the limit of sensitivity of the apparatus. It should be borne in mind that each peak amplitude reading represents the highest peak in a one-eighth second interval.

COMMENTS ON THE DATA

The measurements give two entirely distinct sets of data: (1) average amplitudes; (2) peak amplitudes. The former are obtained by integrating the amplitudes over a period of 15 seconds. This type of average probably has no direct bearing upon the performance of the apparatus intervening between the sound source and the ear. So long an interval as 15 seconds is directly significant only for such effects as heating of circuit elements, breathing and packing of carbon transmitters, etc. unimportant in systems now used for the reproduction of music. These long averages were taken chiefly because they furnished an easy means of checking up on the constancy of a musician's performance as he repeated the same selection over and over again while the several frequency bands were being gone over.

The peak data are taken so that each individual measurement gives the maximum amplitude occurring in a $1/8$ -second interval. This interval approximately represents the duration of a short musical note. In this sense it is an element of music similar to the syllable being an element of speech. For string and wind instruments the peak amplitude measured usually is accompanied by many others of substantially equal

amplitude occurring in the same $1/8$ -second interval. On the other hand, for the most important percussion instruments (drums and cymbals) the sound wave is highly damped and the peak measured probably is decidedly higher than the other peaks occurring throughout that particular $1/8$ -second interval. It is thought that in all cases the peaks as here measured represent a quantity which is individually perceived by a musical ear. If this is correct the overload capacity of apparatus carrying music must be based on these peak quantities.

The frequency analyses for individual instruments agree with those obtained on orchestras in indicating that the sound spectra generated extend over the entire audio range, with components (peak values, of course) in any part of that range so large that if present alone they would be well above the threshold for people with normal hearing. This certainly applies to a person situated in the conductor's usual position. It is not clear that the above components, say in the very highest frequency range, 10,000 to 15,000 c.p.s., are still very audible in the presence of the much larger low frequency components. The answer to this depends on the particular masking conditions which obtain in orchestral music. Further, not enough is known about the simultaneity of occurrence of weak and strong components situated in different frequency regions. The results of listening tests indicate that the normal ear appreciates the contributions made to orchestral music by all the frequency bands used.

The quantities directly measured were electrical only. They are of general interest in so far as they permit the deduction of corresponding acoustic quantities. The total sound output of an instrument or of an orchestra probably is of more interest than any other single quantity. The conversion from electric to acoustic values involves much uncertainty, and the few figures given below must be viewed as rough estimates. The causes of this uncertainty are discussed above, and are briefly mentioned here. From the voltage output of the transmitter the pressure in the sound field can be computed with the aid of the "field" calibration of the instrument. But this is definite only provided the angle of incidence of the sound wave upon the transmitter is known. In general, due to reflections, sound arrives at the transmitter from many directions and with unknown relative amplitudes. The measures taken to obtain pressure values as they would exist in a direct wave from the instrument, have been explained above. They are the first source of uncertainty. Secondly, it is necessary to pass from pressure to power. The computation is straightforward only when reflected waves are neg-

ligible. For other cases it is necessary to make a further assumption as to the relation between potential and total energy, and the absorption coefficients of the room must be known. Thirdly, the measurements are made at one place in the room, and it is necessary to estimate the power distribution in the rest of the space. Finally, there are many minor un-

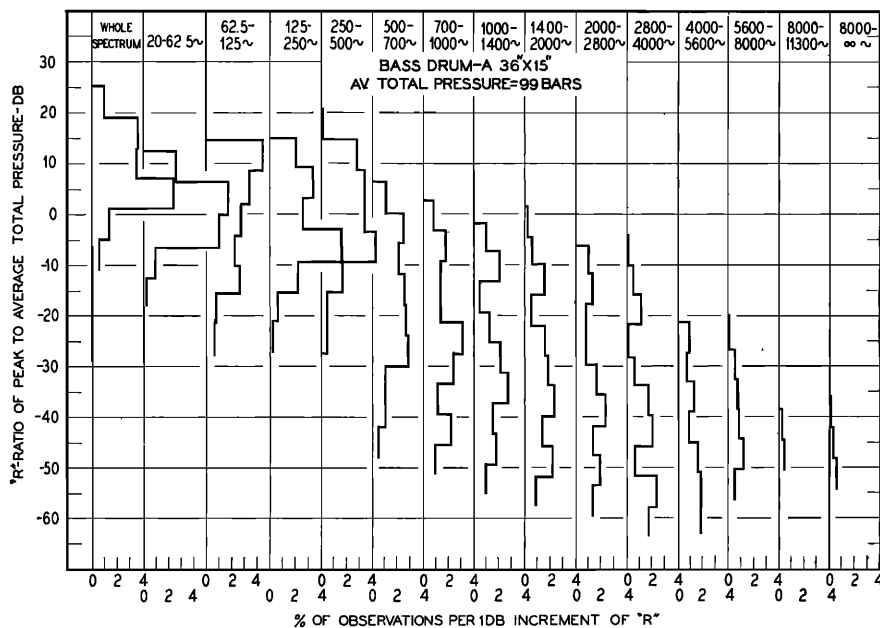


FIG. 33.

certainties, such as the effect upon peak values of the unknown phase relations of the waves arriving from different directions.

With these reservations we proceed to a discussion of the peak powers obtained from the different instruments and orchestras. The results are also shown in tabular form in Fig. 59.

The 36"×15" Bass Drum (A). The microphone is on the drum axis, in front of and 3 ft. distant from the side played. It is seen from Fig. 33 that 6 per cent of the peaks in the "whole spectrum" are between 900 and 1800 bars pressure. Assume the radiation to be confined to a 3 ft. diameter cylinder on one side of the drum. Then the corresponding total sound powers lie between 13 and 51 watts. The logarithmic mean is 25 watts. It is thought that a figure of 20 watts for peak power output of such a drum is low rather than high. Assuming a loud speaker efficiency of 25 percent, the power supplied to it from the last amplifier stage

would be 80 watts. If the drum were playing with a large orchestra and the reproducing system had a fairly uniform response up to about 10,000 c.p.s., the permissible ratio of d.c. input to the a.c. output of amplifier probably is not less than 40:1. Excluding arrangements for balancing out tube distortion which as yet are not in commercial use, the above ratio is thought to be low. Assuming it, the last amplifier stage requires

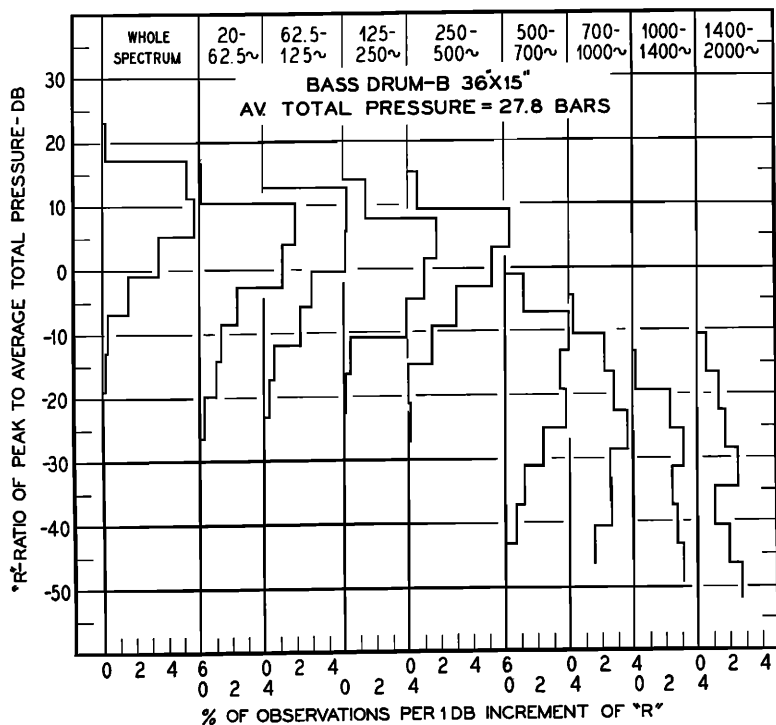


FIG. 34.

a 3.2 kilowatt d.c. input. Of course, this refers to instantaneous power and to a fortissimo manner of playing.

This instrument gives the highest peaks of all single instruments studied. In frequency, it is strong chiefly in the range below 500 cycles, no other instrument except the pipe organ approaching it in this range. In the band between 250 and 500 cycles, 1 percent of the peaks are found to rise to about 10 watts.

The 36" × 15" Bass Drum (B). (Fig. 34) The player and the drum are different, the drum dimensions are the same as for case (A). Estimating peak power on the same basis as above, the value of 1.2 watts is arrived

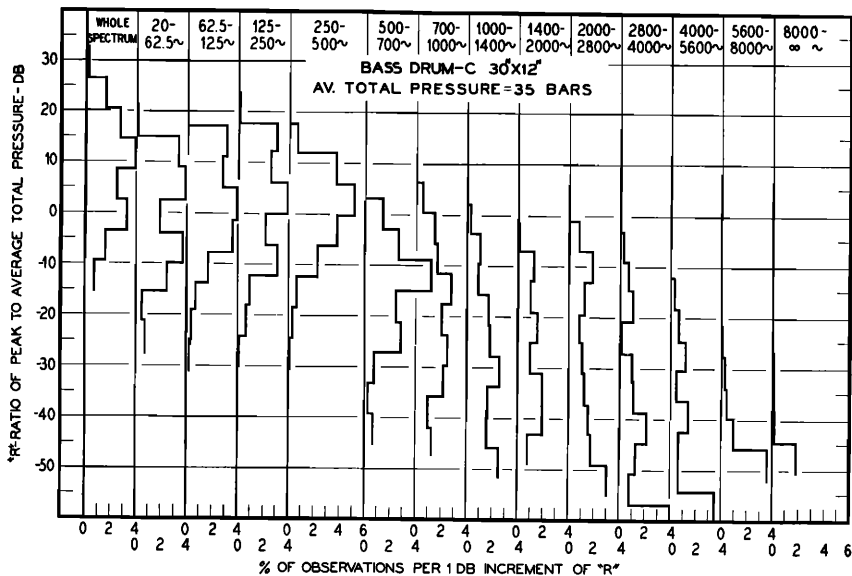


FIG. 35.

at. This assumes the highest peaks to rise to the middle of the highest amplitude zone shown on the curve.

The 30"×12" Bass Drum (C). (Fig. 35) The same player as for drum (A). The peak power output, estimated similarly to (A) and (B), is found to be about 13 watts.

The 34"×19" Bass Drum (D). (Fig. 36) The player is the same as

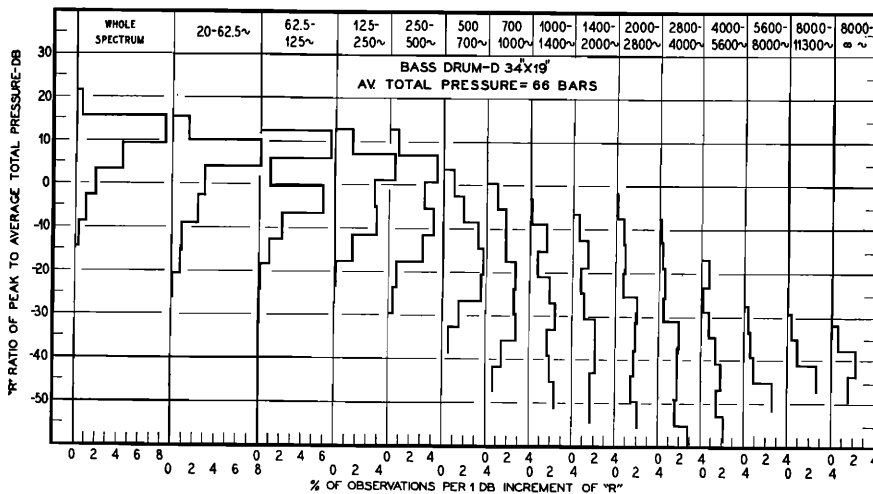


FIG. 36.

with (A) and (C). The peak power is estimated at 5 watts, these peaks occurring during 3 percent of the intervals. In frequency, the highest peaks with this drum are found in the lowest band (20–62.5 c.p.s.), though the other bands below 500 c.p.s. are almost as strong.

Snare Drum. (Fig. 37) The distance is 4 ft., with the microphone located nearly 90° off the axis of the drumhead. Reflections in the room

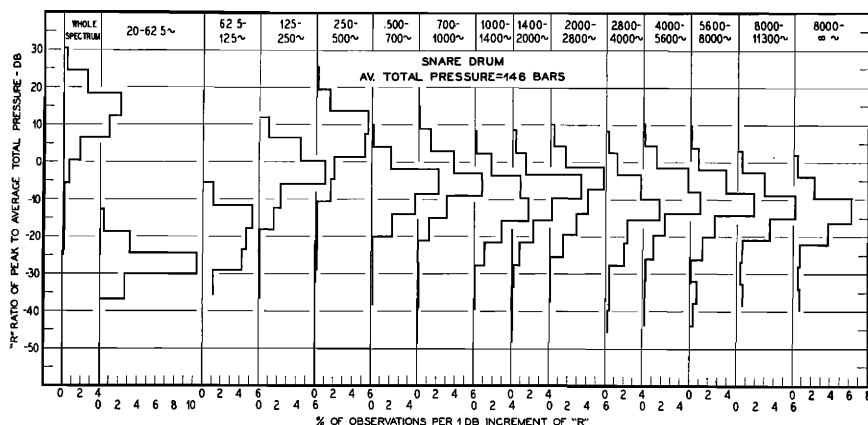


FIG 37.

have an appreciable effect under these conditions. Measurements made with a complex source in the same room (see above) show that the average total pressure falls off 8.5 db. in going from 1 ft. to 4 ft. That at 1 ft. may be safely assumed to be due to the direct wave. Taking this into account, and assuming the radiation to be distributed over a hemisphere, the peak power for the peak pressure of 346 bars is 12 watts. On account of the position of the microphone and the directive properties of the drumhead, this is probably an underestimate.

15" Cymbals. (Fig. 38) The distance is 3 ft., at which the total average pressure has been found to fall 7.2 db. below that at 1 ft. Again assuming distribution of radiation over a hemisphere, the peak power turns out to be 9.5 watts. This is only 4 db. below the peak for the 36"×15" bass drum. The spectrum is particularly rich in high frequencies, the highest peaks lying above 8000 c.p.s. It is also stronger than other instruments studied in all bands above 500 c.p.s.

Triangle. (Fig. 39) Calculating on the same basis as before, the peak power is 50 milliwatts in 1 percent of the intervals, and 12 milliwatts is 37 percent. The high frequencies are prominent, but much lower in absolute value than the cymbals. Six percent of the intervals have peaks

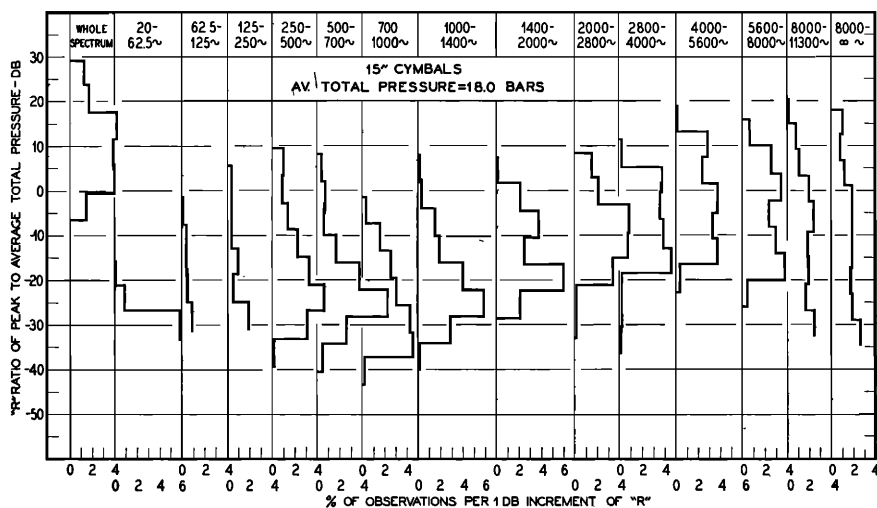


FIG. 38.

of 17 milliwatts in the 5600–8000 cycle band. The very low pressures in some of the lower bands are explained by the fact that the instrument gives a single tone with harmonics.

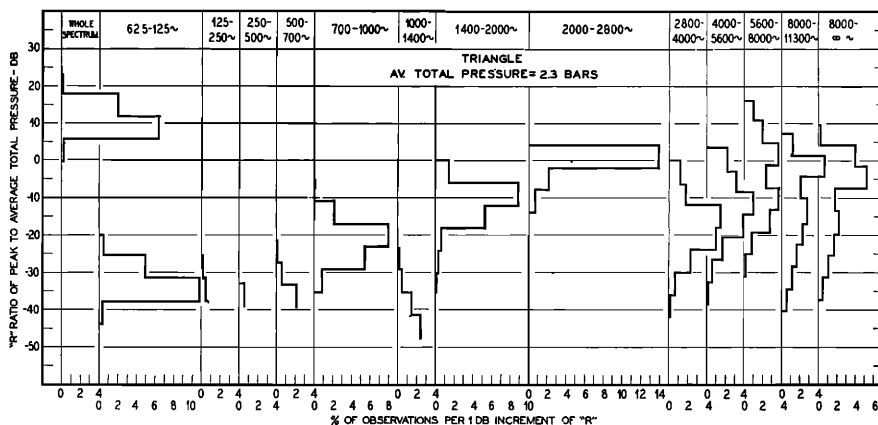


FIG. 39.

In this case, and others following, it may be thought that the assumption of hemispherical distribution of radiation is too conservative. It is thought best to use it, however, since the player's body does act to some extent as a reflector on one side of the instrument, and since it is not known to just what extent phase distortion in the measuring apparatus may change the maximum height of peaks.

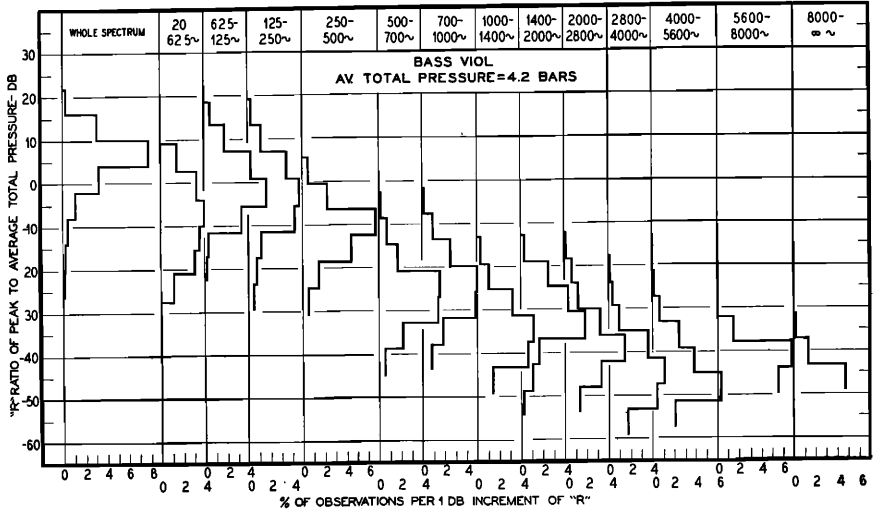


FIG. 40.

Bass Viol. (Fig. 40) Here the instrument is large enough to have pronounced directive properties at the higher frequencies. Since, however,

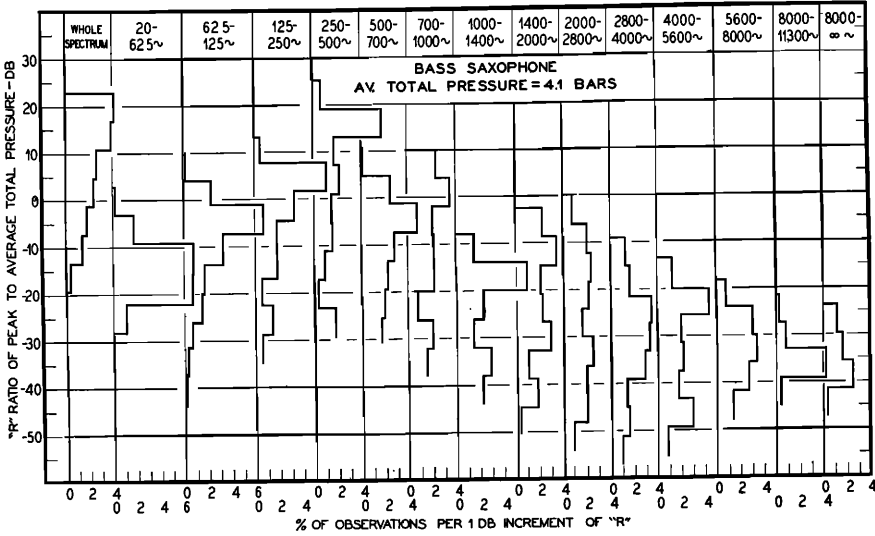


FIG. 41.

most of the output lies below 300 c.p.s., the assumption of uniform distribution over a hemisphere is probably safe. This leads to 156 milliwatts peak power. In the two bands between 62.5 and 250 c.p.s. the peaks rise to 78 milliwatts.

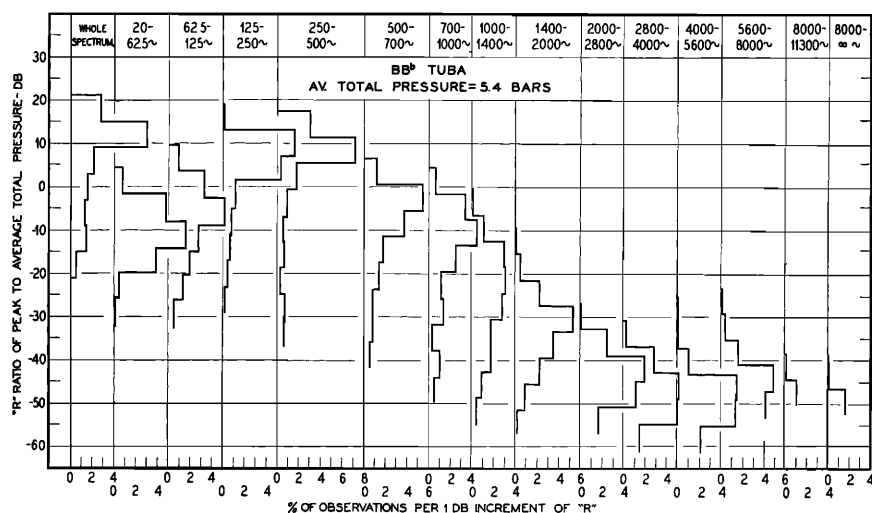


FIG. 42.

Bass Saxophone. (Fig. 41) Twenty-five percent of the intervals in this case give a peak power of 288 milliwatts, with the 250–500 cycle band strongest. Four percent of the intervals in this band have peaks of 228 milliwatts.

BB^b Tuba. (Fig. 42) Here 17 percent of the intervals have peaks of 206 milliwatts. The 250–500 cycle band is strongest, with peaks of 82 milliwatts. The bell of the horn is large and has directive properties. Calculations are made on the basis of the falling off in pressure with distance observed in a complex source using a horn of similar size.

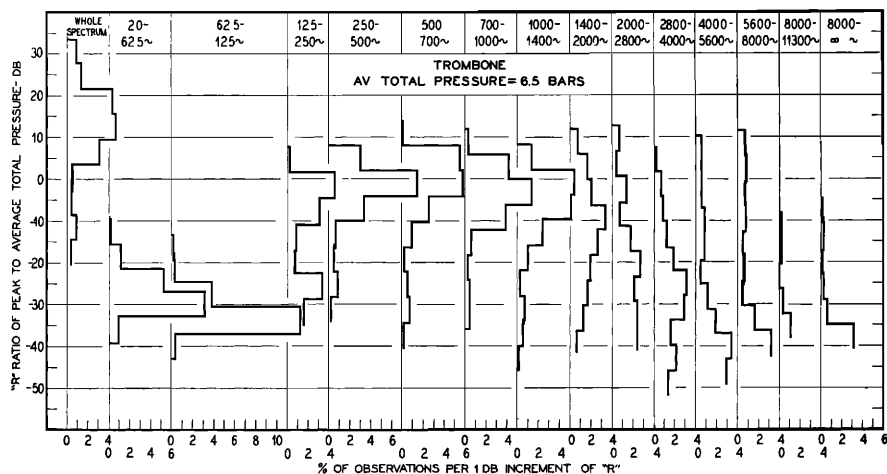


FIG. 43.

Trombone. (Fig. 43) Here, also, a horn of similar size was used with a constant complex source, to determine the rate at which pressure decreases with distance. The calculation leads to a peak power of 6.4 watts for the trombone, occurring during 5 percent of the intervals. If this be compared with the tuba, it is found that the difference of 15 db.

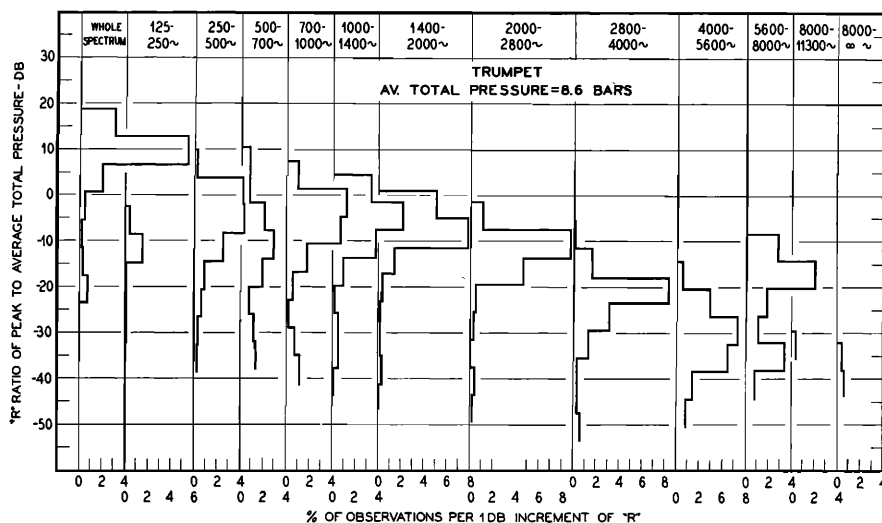


FIG. 44.

lies in a difference of average pressures only to the extent of 2 db., the other 13 db. being due to a difference in peak factor. The highest peaks of the trombone were 31 db. above the average pressure, and this is a higher peak factor than has been found for any other instrument. It is undoubtedly due to the fact that many notes were "attacked" in a sharp explosive manner. The many high peaks observed at high frequencies, even above 8000 c.p.s., are also due to this method of attack. This latter point was directly observed by watching the performance of the peak amplitude meter at high frequencies, while listening at the same time to the playing of the instrument. That the high peaks in the "whole spectrum" are due to the cooperation of peaks of several frequencies is shown by the fact that no single band gives peaks closer than 20 db. to the highest of the whole spectrum. The 500-700 band and the 2000-2800 band are of almost equal strength, but the latter has a higher percentage of peaks at about 60 milliwatts. The player probably used the sharp attack to a greater extent than would ordinarily be the case, so that the sample of trombone music given may be extreme, both in

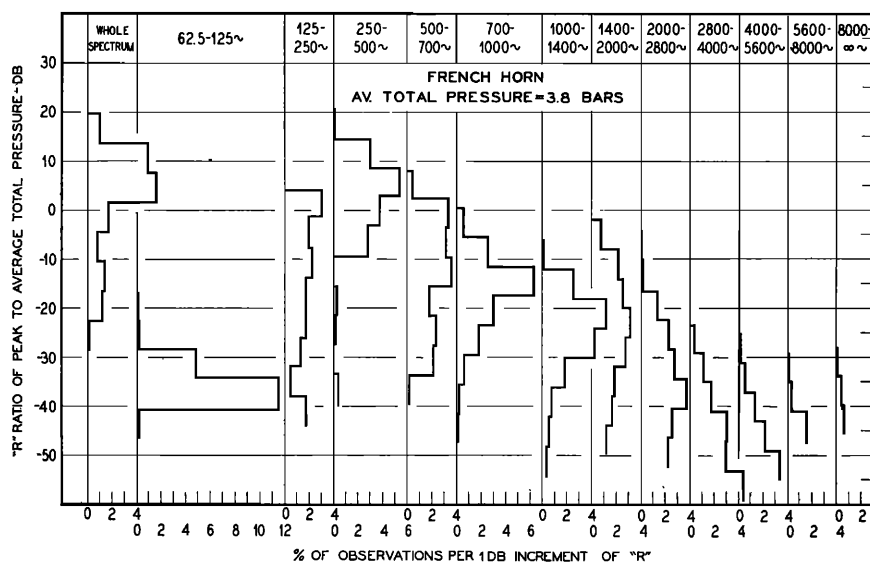


FIG. 45.

height of peaks and in frequency range; yet these figures must be considered if all types of playing are to be perfectly reproduced.

Trumpet. (Fig. 44) The peak power is 314 milliwatts, slightly higher than bass saxophone and tuba. Eighteen percent of the intervals give peaks of this magnitude. The two bands between 250 and 700 c.p.s. both have peaks of about 47 milliwatts.

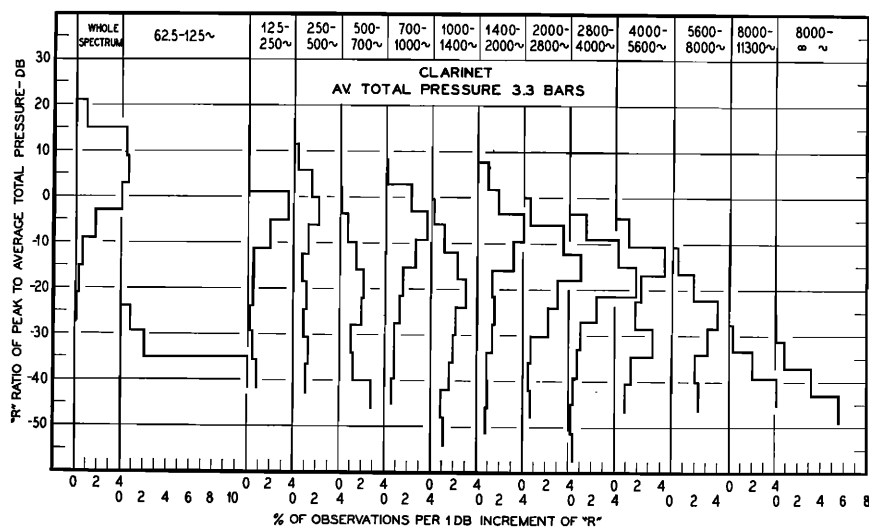


FIG. 46.

French Horn. (Fig. 45) The peak power is 53 milliwatts, in 6 percent of the intervals. The 250–500 c.p.s. band is very much the strongest, having peaks as high as those in the whole spectrum, but during only 1 percent of the intervals.

Clarinet. (Fig. 46) The peak power for the whole spectrum is about the same as for the French horn,—50 milliwatts. The strongest band

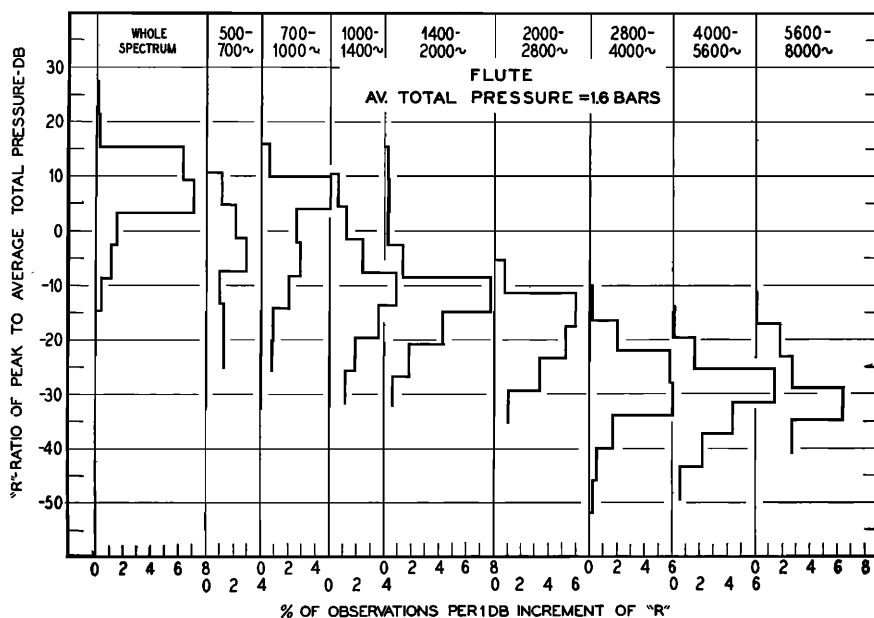


FIG. 47.

here is also the 250–500 c.p.s. band, but the peaks in it are 10 db. below those in the whole spectrum. The high frequencies are more pronounced, however, than with the French horn.

Flute. (Fig. 47) The peaks in 1 percent of the intervals rise to 55 milliwatts, in 1.5 percent to 14 mw., and in 38 percent to 3.5 mw. In frequency, the 700–1000 and 1400–2000 c.p.s. bands are about equally strong, with 4.5 mw. Some of the lowest and highest bands were omitted, but qualitative observations showed that the omitted bands were weaker than those shown.

Piccolo. (Fig. 48) About 1/2 percent of peaks rise to 84 milliwatts, while 10 percent are about 21 mw. The strongest band is 2000–2800 c.p.s., with 3 percent of intervals giving 21 mw.

Piano (A). (Fig. 49) The nature of the source makes it more difficult

to calculate total acoustic power from the observed pressures. Three methods are given.

First Method: The microphone is 10 ft. from the piano, and the room is comparatively small, so that it may be assumed that the sound pressures measured are those of completely diffused sound in the room. The reverberation time of the room is known, and the power may therefore

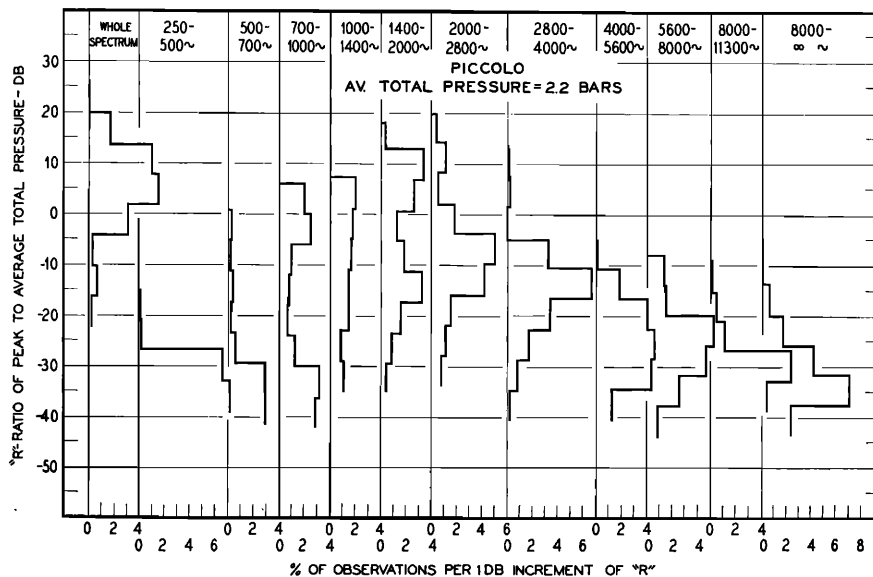


FIG. 48.

be calculated. The peak power obtained in this manner is 166 milliwatts.

Second Method: A single measurement with a complex point source, with the microphone 12 ft. away and turned 90° from the direction of the source, showed that the average pressure was 14.6 db. below that with the source at one ft., directly in front of the transmitter. If we assume the former position corresponds to completely diffused sound, and the latter to a direct wave, it is easily shown that the pressure of diffused sound in the room is the same as if the entire radiation were distributed over a sphere with a radius of 5.37 ft. These assumptions are probably not far from right. Making the calculation for the piano on this basis, the peak power is 437 milliwatts. This is probably high, since even at ten feet the directive properties of the raised lid of the piano may cause a greater concentration of sound than would be found in completely diffused radiation.

Third Method: Auxiliary piano measurements have been made using a part of a certain mechanical roll. The microphone was placed in the same position as with the more general measurements, and again in a position very close to the open piano. The average pressure was 9.6 db. less in the more distant position. The closer position was in the opening of the dihedral angle between sounding board and raised lid, where it

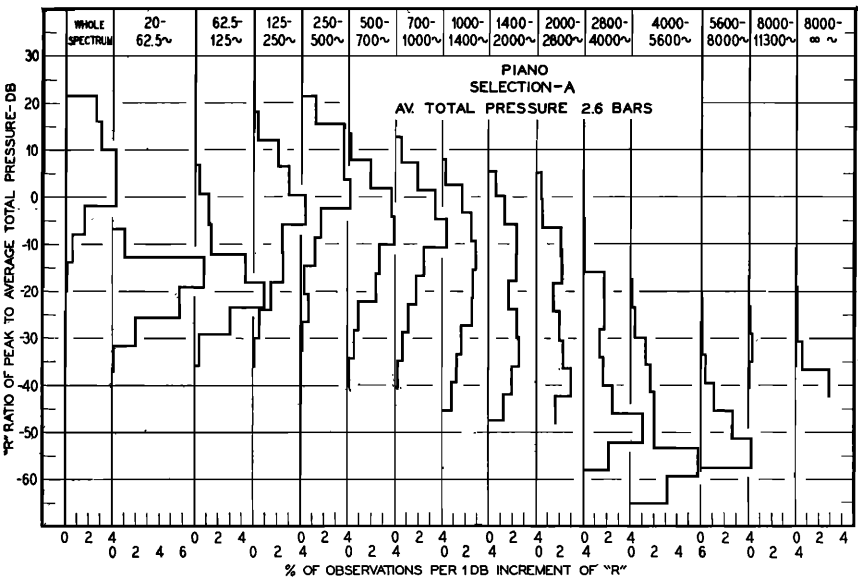


FIG. 49.

might be assumed the sound was distributed over an area roughly $3' \times 6'$ in size. Calculating the peak power on this basis gives 198 milliwatts.

The remarkable thing about these three results is their agreement, rather than their disagreement, in view of the assumptions involved. An average of the three gives 267 milliwatts. The 250-500 c.p.s. band has the highest peaks, of practically the same magnitude as those of the whole spectrum. The instrument is a high grade grand piano with a reproducing mechanism, and the selection in this case is Liszt's "Second Hungarian Rhapsody."

The selection being played mechanically, the question arises as to wheather peaks are as high as an artist can produce on the same instrument. A special roll was prepared, and an artist employed. The results showed that he could strike single notes with an average peak 9 db. higher than the highest possible with the roll, but that when he played chords involving a number of notes, his peaks were about 4 db. lower

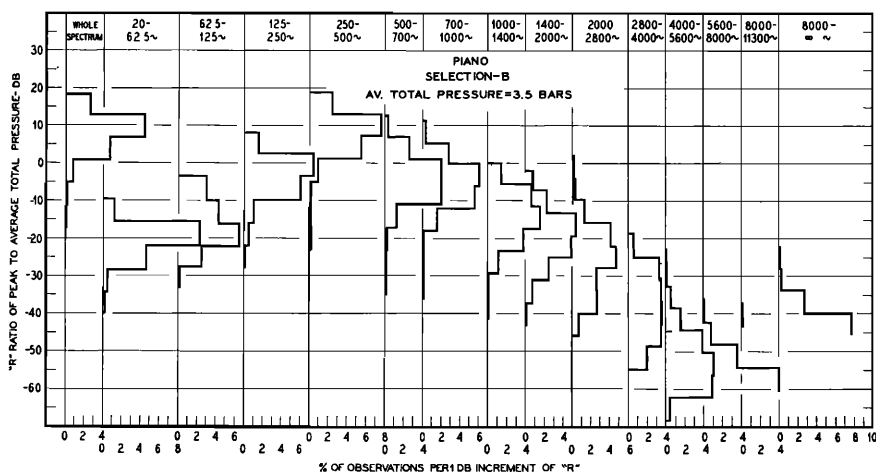


FIG. 50.

than those with the roll. The highest possible peaks on this piano, with an artist playing, would seem to be about 2 watts.

Piano (B). (Fig. 50) The same piano is used, but with a different roll,

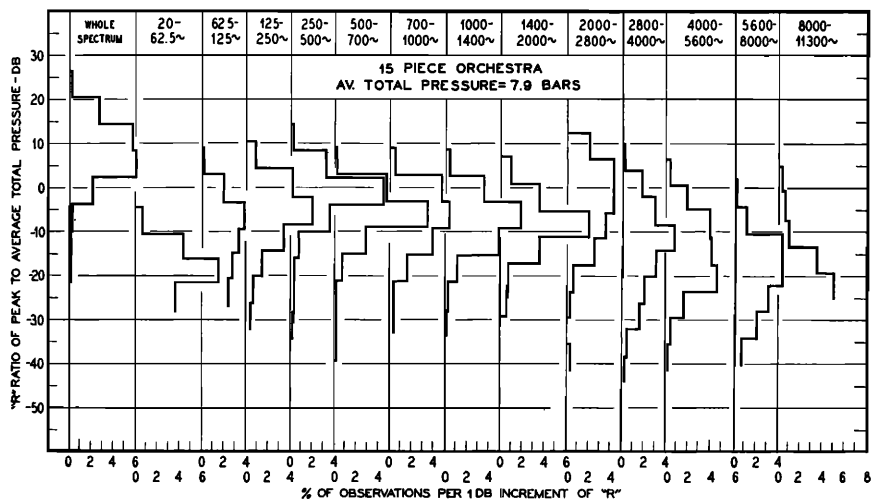


FIG. 51.

in this case a four-handed jazz record. The results, as far as peak power is concerned, are about the same as in (A), the average of the three methods giving 248 milliwatts. Again the 250–500 c.p.s. band has peaks as high as those in the whole spectrum.

Fifteen-Piece Orchestra in Laboratory. (Fig. 51) The room is the same

as that for the piano measurements. The first and second methods used with the piano may be applied here, except that with Method 1 the added absorption due to the presence of fifteen men is taken into account. Taking the peak pressure as 125 bars, the peak power by the 1st method is 5.3 watts. By the second method we get 12.7 watts. The

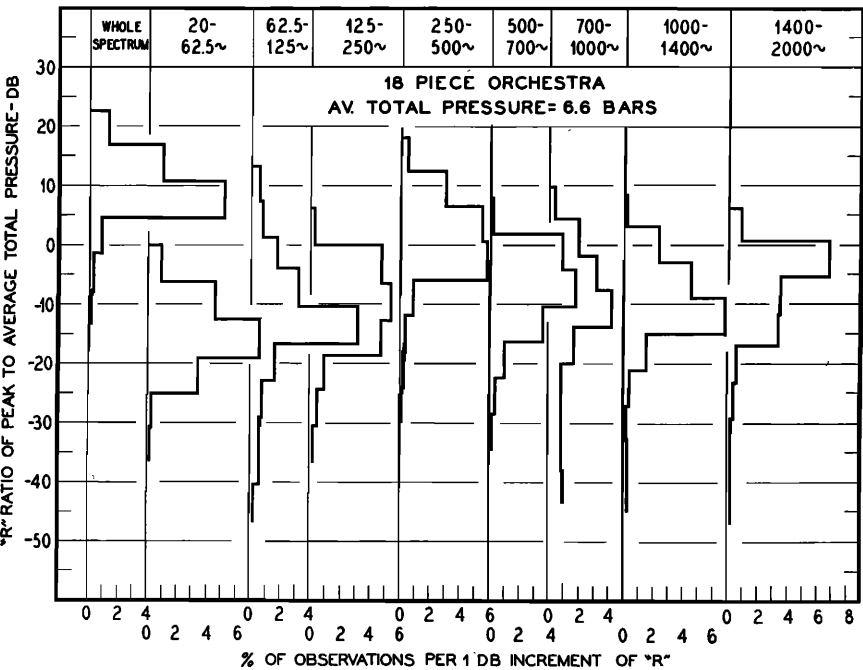


FIG. 52.

agreement, as before, is thought to be good. The average is 9 watts. No doubt the same orchestra, playing in a larger room, would do so at a distinctly higher level. In frequency, the 250–500 c.p.s. band has peaks of 450 milliwatts in 1.5 percent of the intervals, while the 2000–2800 c.p.s. band has peaks of 320 milliwatts in 12 percent of the intervals. The bands between these are not much lower.

Eighteen-Piece Orchestra in Laboratory. (Fig. 52) The peak power by Methods 1 and 2 is 1.5 watts and 3.5 watts, respectively. The average is 2.5 watts. This is 5.5 db. below that found for the fifteen-piece orchestra, but it occurs during 8 percent of the intervals, as compared with 1.5 percent for the other. The orchestra did not play long enough to permit the completion of the frequency spectrum, but of those bands taken, 250–500 c.p.s. is strongest with 800 milliwatts.

Seventy-Five-Piece Orchestra in Theater (A). (Fig. 53) The difficulties connected with measurements made on sound picked up in the theater have already been mentioned. They are less certain than measurements made entirely in the laboratory, chiefly because of the added amplifiers and the long line between theater and laboratory. In addition, no data on reverberation time, or on the decrease of sound pressure with dis-

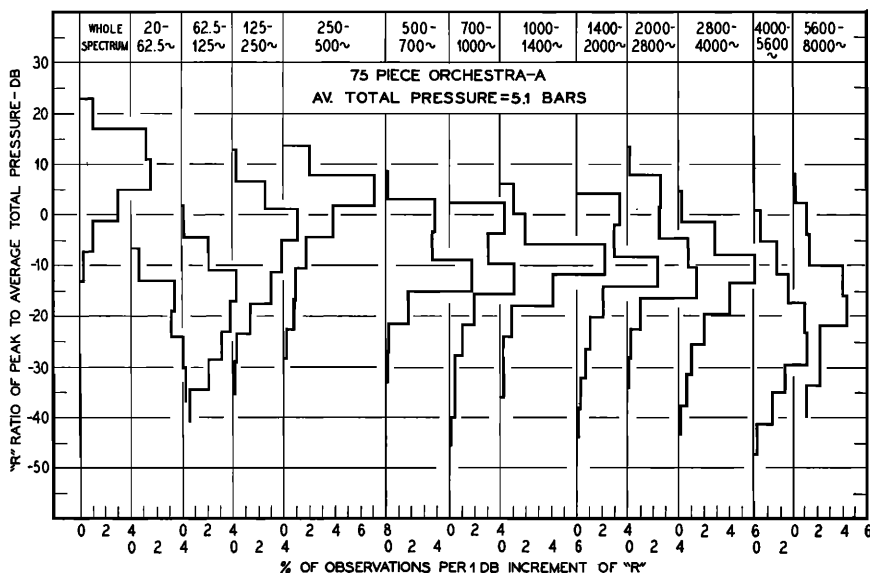


FIG. 53.

tance, are available for measurements taken in the theater. Power calculations are based on the following considerations. The players are distributed over about 210° of a circle approximately 50 ft. in diameter. The microphone is mounted near the conductor's desk, approximately in the center of the circle. It is reasonable to take the direct sound waves as predominating. The assignment of a single distance between microphone and source is impossible in this case. Considering the location of the various instruments, an effective distance of about 15 ft. seems low rather than high. Assuming a uniform distribution over a hemisphere of 15 ft. radius, the peak power is found to be 8.2 watts. The three bands 125–250, 250–500, and 2000–2800 c.p.s. are nearly equally strong, with peak powers of about 1 watt. The music in this case is of a lively nature, with brasses prominent, and a moderate amount of drums. Two very similar selections are included. One of them is shown in bands Nos. 1, 3, 5, 7, 9 and 11 (counting the 20–62.5 c.p.s. band as 1); the other selection

is shown in the "whole spectrum" and in bands Nos. 2, 4, 6, 8, 10 and 12. No readings were taken above 8000 c.p.s.

Seventy-Five-Piece Orchestra in Theater (B). (Fig. 54) the music here is that of an overture of about ten minutes length. Extremes of volume are wider than with (A). There are parts in which strings or wood-wind predominate, but also instants where brass, drums and cymbals are used. Three selections are used, all of them fitting the above description. The first is an arrangement of Bizet music, and is shown in the first "whole

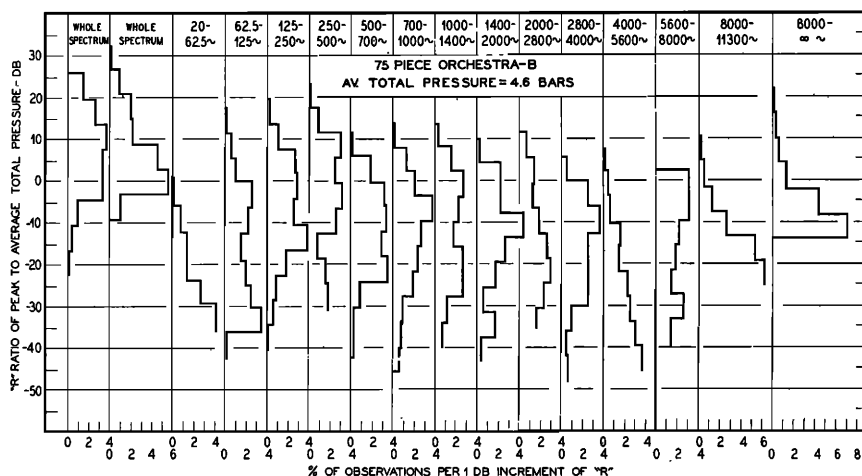


FIG. 54.

spectrum," and in bands 1, 3, 5, 8, 10, and 12. The second selection is the overture to "Mignon," and is shown in the second "whole spectrum," and in bands 2, 4, 6, 7, 9, and 11. The third selection is an arrangement of Puccini music, and is shown only in the two bands above 8000 c.p.s. Average pressures in the three selections fall within a range of 1 db., so that the single average of 4.6 bars may be used in all cases without serious error. This is mentioned particularly to prevent doubts as to the absolute values of peaks in the bands above 8000 c.p.s. Power calculations in the first "whole spectrum" show peaks of 13 watts in 9 percent of the intervals and in the second peaks of 66 watts in 1 percent of the intervals. The 250-500 c.p.s. band is highest with peaks of 6.7 watts, and the 8000- ∞ band next with a small percentage of peaks at 5.3 watts. These high frequency peaks are due to a small percentage of intervals in which the cymbals were very prominent.

Seventy-Five-Piece Orchestra in Theater (C). (Fig. 55) This is ballet music, accompanying a scene called "Autumn Leaves." Two parts of

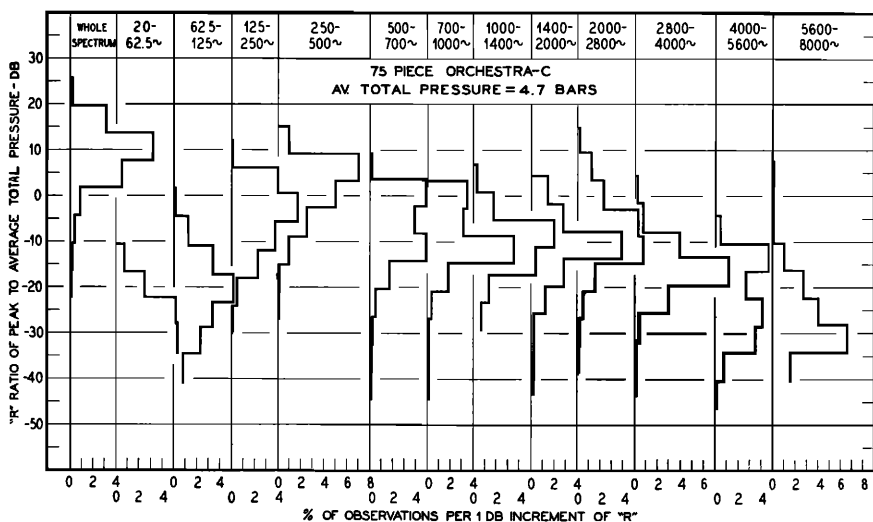


FIG. 55.

the same selection are used, the first part appearing in the whole spectrum and in bands 1, 4, 6, 7, 9 and 11, the second part in bands 2, 3, 5, 8, 10 and 12. The bands above 8000 c.p.s. are omitted. The peak power is 14 watts, with both 250–500 and 2000–2800 c.p.s. bands having peaks of 1.4 watts.

Seventy-Five-Piece Orchestra in Theater (D). (Fig. 56) The two bands below 125 c.p.s. and the four bands above 4000 c.p.s. are for a selection

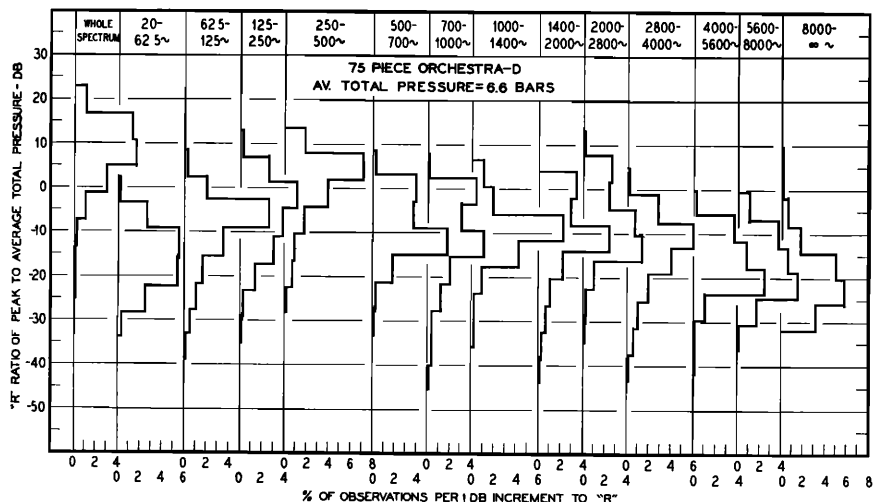


FIG. 56.

very strong in bass drums and cymbals. The intermediate bands, and the whole spectrum, are the same as in (A). It may be noted, by comparing these curves with those of (A), how both ends of the spectrum are raised by the strong prominence of bass drums and cymbals.

Theater Pipe Organ (A). (Fig. 57) The organ is about 60 ft. long. The microphone is centrally in front of it, about 12 ft. from the middle. A

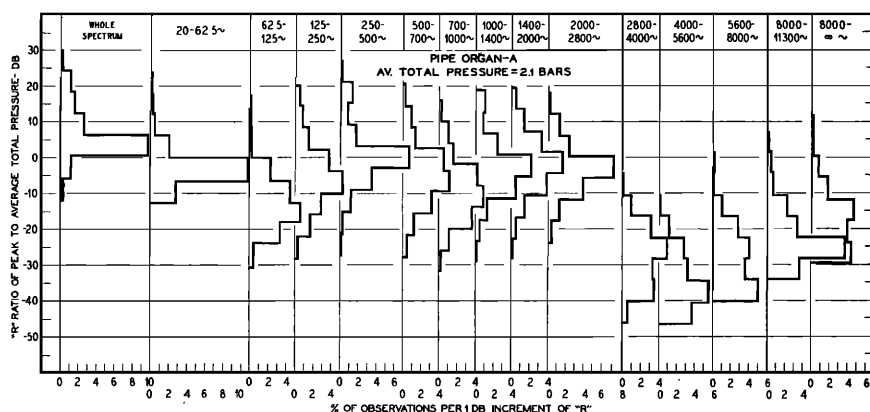


FIG. 57.

semi-circular reflecting wall about 10 ft. high is back of the transmitter, its nearest point about 12 ft. from the transmitter. Obviously the situation is even more uncertain than for the orchestra. As a guess, an effective distance of 15 ft., and uniform radiation over a quarter of a sphere are assumed. This gives a peak power of about 3.5 watts. The 250–500 c.p.s. band is strongest, with peaks of about 1.7 watts. The selection is Kreisler's "Liebesleid," modified so as to make use of the different classes of pipes in the organ, and to include a wide range of volume.

Theater Pipe Organ (B). (Fig. 58) The selection is "Hail, Columbia" played fortissimo throughout. The full organ is used, with swells open wide. Several 16 ft. pipes are included, and also "resultant 32's," the latter being combinations which give difference tones corresponding to the tone of an open pipe 32 ft. long. The purpose was to obtain extreme values for the "whole spectrum" and for the two ends of the frequency range. Measurements were made in the whole spectrum, in three bands at each end of the spectrum, and in one intermediate band, 1000–1400 c.p.s. The calculations indicate that 36 percent of the intervals have peaks of 12.6 watts. In the bands, the lowest is strongest, with a few peaks at 10 watts, and many at 2.5 watts. The results would indicate

that the organ is not quite as strong as the strongest bass drum tested, on the basis of peaks in the whole spectrum; but that when played at full volume, the peaks below 62.5 c.p.s. are quite as strong as those of the drum.

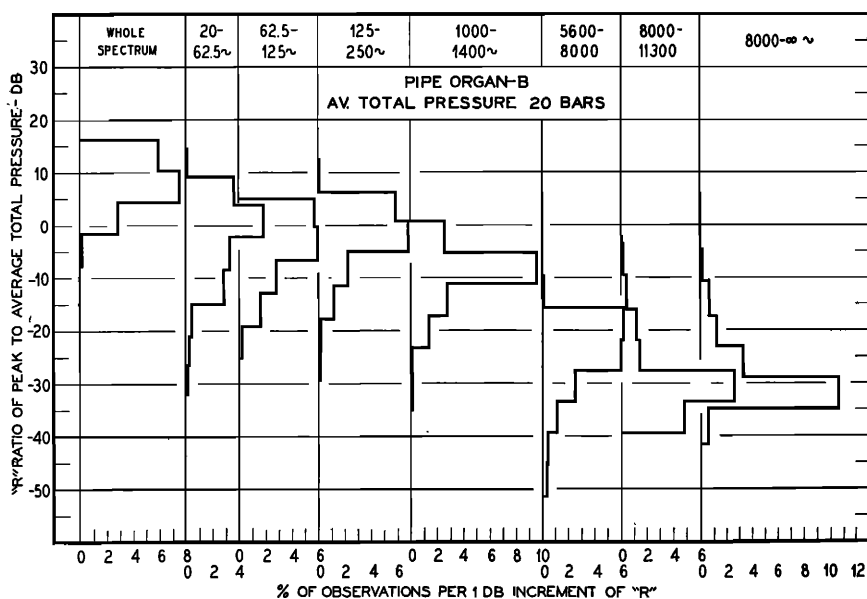


FIG. 58.

LOWEST LEVELS IN MUSIC

In all of the foregoing, the upper limit of amplitude has been sought. The lowest level that it may be necessary to reproduce is also important because of the bearing it may have on permissible extraneous noises. Some data are available on this point, though they are very incomplete.

The highest pressures obtained in the laboratory were from a 36" × 15" bass drum. It was thought that the lowest level it might be desirable to reproduce would be that of a violin played very softly. A violin player was obtained and placed at a distance of 3 ft. from the transmitter, the same distance as that used with the drum. He was told to play at the lowest level that could be used with an audience. The resulting average pressure in the whole spectrum, in a 15 second interval of uniform playing, was 0.52 bars. The highest average pressure obtained from the drum in an equal interval was 133 bars. The range is 48 db. This is not sufficient, however, for reproduction must go at least as low as the lowest average amplitude of the violin, and at least as high as the highest

peak amplitude of the drum. The highest peaks found with the drum were about 1250 bars, and the range is increased to 68 db. There still remains doubt as to whether the two music samples chosen were extremes.

PEAK POWER OF MUSICAL INSTRUMENTS					
INSTRUMENT	WHOLE SPECTRUM PEAK POWER WATTS	CORRESPONDING % OF INTERVALS	BAND CONTAINING MAXIMUM PEAKS	BAND PEAK POWER WATTS	CORRESPONDING % OF INTERVALS
36"×15" BASS DRUM - A	24.6	6	250-500 c.p.s.	9.8	1
36"×15" BASS DRUM - B	1.2	1½	20-62.5	0.24	1
			250-500	0.19	4
30"×12" BASS DRUM - C	13.4	1	125-250	1.7	1
34"×18" BASS DRUM - D	4.9	3	20-62.5	1.2	9
SNARE DRUM	11.9	2½	250-500	3.7	1
15" CYMBALS	9.5	7½	8000-11300	0.95	1
TRIANGLE	0.050	1	5600-8000	0.017	6
	0.012	37			
BASS VIOL	0.156	2	62.5-125	0.078	3
			125-250	0.078	2
BASS SAXOPHONE	0.268	25	250-500	0.228	4
BB ^b TUBA	0.206	17	250-500	0.082	18
TROMBONE	6.4	5	500-700	0.064	1
			2000-2800	0.051	4
TRUMPET	0.314	18	250-500	0.047	1½
			500-700	0.047	4½
FRENCH HORN	0.053	6	250-500	0.053	1
				0.013	18
CLARINET	0.050	5½	250-500	0.0055	2½
FLUTE	0.055	1	700-1000	0.0045	4
	0.014	1½			
	0.0035	38	1400-2000	0.0045	2
PICCOLO	0.084	½			
	0.021	10	2000-2800	0.021	3
	0.166	16	250-500	0.166	7
PIANO-A	0.437	16	250-500	0.437	7
	0.198	16	250-500	0.198	7
	0.267	16	250-500	0.267	7
PIANO-B - AVERAGE	0.248	16	250-500	0.248	14
15-PIECE ORCHESTRA	9.0	1½	250-500	0.45	1½
AV. OF 2 METHODS	2.2	16	2000-2800	0.32	12
18-PIECE ORCHESTRA	2.5	8	250-500	0.80	3
AV. OF 2 METHODS			2000-2800	NOT TAKEN	
75-PIECE ORCHESTRA A	8.2	6	125-250	0.82	2
			250-500	1.03	12
			2000-2800	1.03	1½
75-PIECE ORCHESTRA B	13.4	9	250-500	6.7	1
	66.5	1	8000-∞	5.3	1
75-PIECE ORCHESTRA C	13.9	1½	250-500	1.4	6
			2000-2800	1.4	1½
75-PIECE ORCHESTRA D	13.8	6	125-250	1.7	2
			250-500	1.7	11
			2000-2800	1.7	1
PIPE ORGAN - A	3.5	1½	250-500	1.75	1
				0.44	8
PIPE ORGAN - B	12.6	36	20-62.5	100	1
				25	22

Fig. 59.

Among all data taken on the theater orchestra, the extremes of pressures have been sought out. The lowest 15-second average pressure obtained was 0.27 bars. This may be high, both because the music was probably not at minimum level during the entire interval, and because a very low fluxmeter reading, such as was obtained, is apt to be in error on the high side rather than the low. The highest peak pressure, obtained with the microphone in the same position relative to the orchestra, was 430 bars. The range is 64 db.

It is concluded that a system reproducing a large orchestra, the music ranging from a fortissimo ensemble to a pianissimo violin solo, might be called upon to handle a range as great as 70 db.

HISTORICAL NOTE

The analysis of instrumental musical sounds is a problem of rather old standing. It has been approached at various times by both physicists and psychologists, since the days of Helmholtz's work with his resonators. A summary of the earlier work and a description of his own notable contributions to the subject may be found in D. C. Miller's book.⁵ The results were largely confined to the determination of the frequencies of those components which the investigators held to be responsible for the musical quality of the instrument. A more recent study along this line has been made by C. Stumpf,⁶ who determined the characteristic frequencies for a number of wind instruments. Still more recently H. M. Browning⁷ and H. Backhaus⁸ have reported on the characteristic partials of the violin.

In the earlier measurements neither the relative nor absolute physical magnitudes of the frequency components were determined. This was not possible until microphones admitting of absolute calibration over the frequency range became available. The condenser microphone, calibrated by the thermophone method, or by some one of several other methods more recently applied,⁹ has proved useful for this purpose. A summary of its application, in conjunction with electric filters, to the statistical analysis of speech sounds, street noise, and piano music, is given in the paper already referred to.¹⁰ J. F. Cellerier¹¹ has used a piezo-

⁵ "Science of Musical Sounds," Macmillan Company, 1916, New York.

⁶ *Zs. f. Phys.*, 38, 9, 10, 1926.

⁷ *Phil. Mag.*, November, 1926.

⁸ *Naturwissenschaften*, October 18, 1929.

⁹ e.g., L. J. Sivian, *Bell System Technical Journal*, January, 1931.

¹⁰ L. J. Sivian, *Bell System Technical Journal* October, 1929.

¹¹ *Comptes Rendus*, January 6, 1930.

electric microphone, calibrated with a thermophone, to analyze the sounds produced by automobile horns. E. Meyer and P. Just¹² have measured the total sound outputs of several string and wind instruments. H. Lueder,¹³ also using a condenser microphone and electric filters, has made a statistical analysis of the amplitudes and spectra of a large number of musical instruments and of orchestras.

¹² Zs. f. tech. Phys., **10**, 8, 1929.

¹³ Wissenschaftliche Veröff. a. d. Siemens-Konzern, May, 1930.